Food Remedy API

RE039: Symptom-to-Additive Lookup Tool

Prepared For: Food Remedy API Team

Author: Riken patel

Date: 15 August 2025 (T2 2025)

Objective:
People often experience unexplained symptoms such as headaches, rashes, or breathing difficulties after eating. The goal of this task is to develop an interactive tool that allows users to select a symptom and discover which food additives may be responsible. The tool links symptom information from medical and allergen sources to a reference list of additives, helping users explore possible dietary triggers.

1. **Symptom-to-Additive Associations**

Based on research from allergy foundations, medical studies, and food safety databases, the following relationships are commonly observed:

Symptom	Potential Additives (E-number / Name)	Common Food Sources	Typical Reaction
Headache / Migraines	E621 – MSG, E220–E228 – Sulfites, Nitrates/Nitrites, E102 – Tartrazine, BHA/BHT	Processed meats, soft drinks, snacks, dried fruits, wines	Headache, flushing, chest tightness
Rash / Hives / Skin	Sulfites, Tartrazine, Benzoates (E210–E219), BHA/BHT, Parabens, Annatto (E160b)	Sauces, condiments, cereals, candies, preserved foods	Hives, rashes, skin irritation
Breathing Trouble / Asthma	Sulfites, Tartrazine, Benzoates, MSG (rare)	Wines, dried fruits, processed foods	Wheezing, shortness of breath, asthma exacerbation
Allergic-like Reactions / Hives	Sulfites, Benzoates, Nitrates/Nitrites, Tartrazine	Soft drinks, processed meats, preserved foods	Hives, swelling, mild allergic reactions

2. Mechanisms of Adverse Reactions

Food additive reactions can be broadly classified into allergic, intolerance, and pseudoallergic responses:

- Allergic Reactions:
  The immune system mistakenly identifies an additive as harmful, producing IgE antibodies. Symptoms can range from mild rashes to severe reactions such as anaphylaxis.
- Intolerance Reactions:
  These are non-immune responses, often dose-dependent. They commonly result in headaches, gastrointestinal issues, or mild skin irritation.
- Pseudoallergic Reactions:
  Certain additives can directly activate mast cells without involving IgE antibodies, causing hives, flushing, or mild respiratory distress.

Understanding these mechanisms helps users and healthcare professionals identify potential triggers more accurately.

3. Tool Design Overview

To maximize usability, the Symptom-to-Additive Lookup Tool incorporates the following key components:

1. Symptom Input Interface:
   Users can select one or multiple symptoms they are experiencing from a dropdown or checklist.
2. Additive Database:
   The system maintains a comprehensive database of additives, including:
   - E-numbers and common names
   - Known symptom associations
   - Typical food sources
   - Recommended intake limits
3. Matching Algorithm:
   An algorithm matches user-selected symptoms to potential additives, ranking them by likelihood and severity of reaction.

4. Information Display:
   Users receive a detailed list of potential additives along with:
   - Foods containing them
   - Common reactions
   - Tips for avoidance or alternative products
5. Educational Resources:
   Links to allergy foundations, food safety databases, and clinical studies provide further reading.

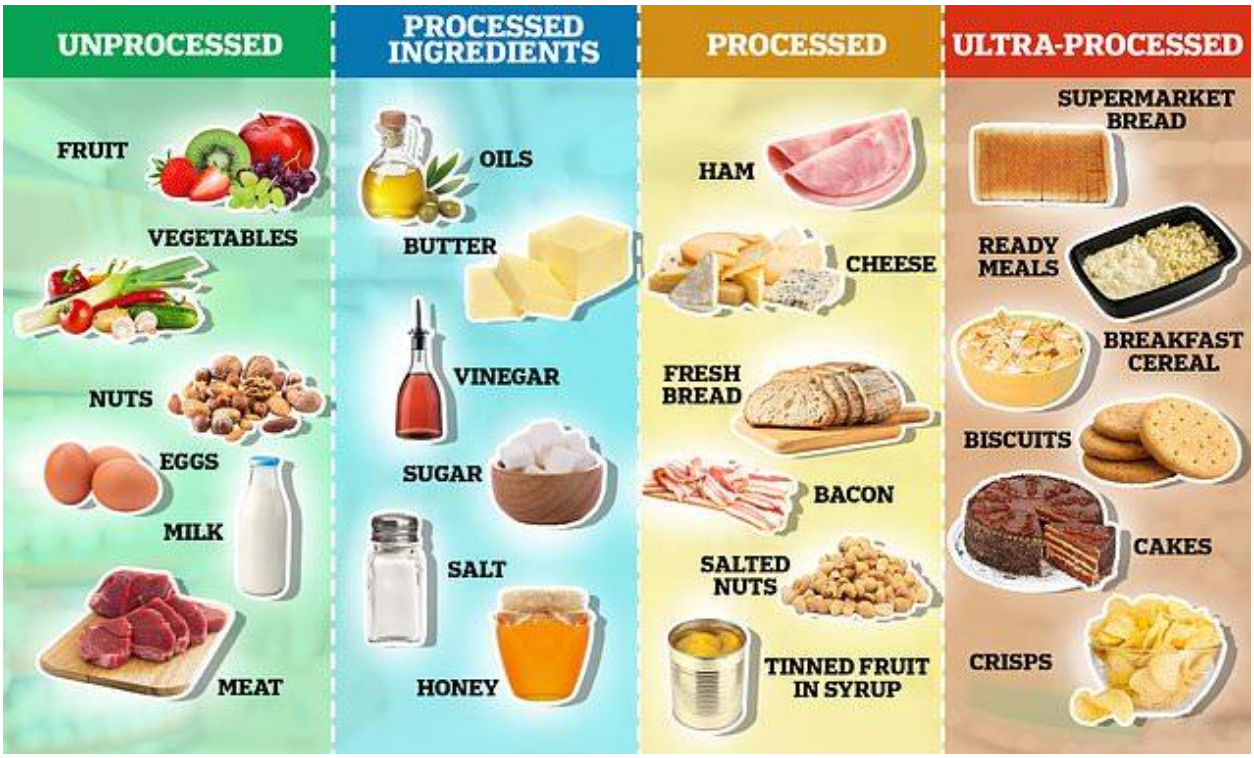
4. Visual Components

Visuals help users understand complex information at a glance. Suggested graphics include:

- **Symptom-Additive Matrix:**
  A color-coded table mapping symptoms to associated additives.

- **E-number Classification Chart:**
  Shows categories like colors (E100–E199), preservatives (E200–E299), antioxidants (E300–E399).

- **Food Sources Infographic:**
  Illustrates which processed foods typically contain certain additives.



5. Benefits of the Tool

- Empowers Users: Individuals can explore potential triggers without guesswork.
- Supports Health Professionals: Provides a quick reference for dietitians or allergy specialists.
- Educational: Raises awareness of additives and their potential health impacts.
- Data-Driven: Combines symptom reporting with evidence from clinical studies and food safety databases.

6. References

Verywell Health, 2025. Allergy to food additives and preservatives. [online] Available at: https://www.verywellhealth.com/allergy-to-food-additives-and-preservatives-82899 [Accessed 15 August 2025].

Sharecare, 2025. Food additive allergies. [online] Available at: https://www.sharecare.com/allergies/food-allergies/food-additive-food-allergy [Accessed 15 August 2025].

Asthma and Allergies, 2025. Adverse reactions to food additives. [online] Available at: https://asthmaandallergies.org/food-allergies/adverse-reactions-to-food-additives/ [Accessed 15 August 2025].

Healthy Food Guide, 2025. Can food additives trigger allergic reactions? [online] Available at: https://www.healthyfood.com/advice/can-food-additives-trigger-allergic-reactions/ [Accessed 15 August 2025].

Food Sensitivity Hub, 2025. Food additive intolerance. [online] Available at: https://foodsensitivityhub.com/food-additive-intolerance/ [Accessed 15 August 2025].